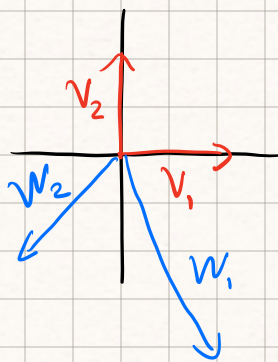


Changing bases in coordinates

$$B_1 = [v_1, v_2]$$

What is v_1 in B_1 coords? $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{B_1}$



$B_2 = [w_1, w_2]$ What are the coords of w_1 in B_2 ?

$$w_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{B_2}$$

What are the coords of w_1, w_2 in B_1 ?

$$w_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}_{B_1} \quad w_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}_{B_1}$$

Is there a matrix (linear transformation) s.t. $M \begin{bmatrix} \\ \end{bmatrix}_{B_2} = \begin{bmatrix} \\ \end{bmatrix}_{B_1}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & b \\ -2 & d \end{bmatrix}}_M \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & b \\ -2 & d \end{bmatrix}}_M \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

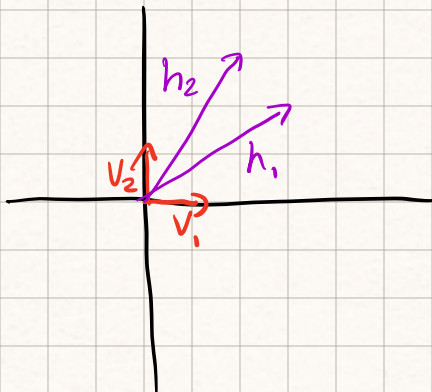
$$M = \begin{bmatrix} 1 & -1 \\ -2 & -1 \end{bmatrix}$$

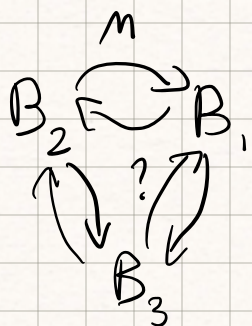
takes B_2 to B_1

test

$$\begin{bmatrix} 1 & -1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{B_2} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}_{B_1} \quad \checkmark$$

$$B_3 = [h_1, h_2]$$



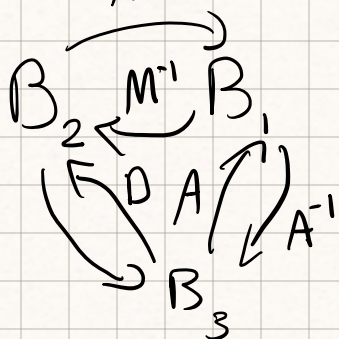


$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix}_{B_3} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}_{B_1}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$a=3, c=2$$

$$\begin{bmatrix} 3 & b \\ 2 & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{B_3} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}_{B_1} \quad A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$



$$M = \begin{bmatrix} 1 & -1 \\ -2 & -1 \end{bmatrix} \quad M^{-1} =$$

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \quad A^{-1} =$$

$$D =$$

$$D^{-1} =$$

$$M^{-1}:$$

$$\begin{array}{cc|cc} 1 & -1 & 0 & 0 \\ -2 & -1 & 0 & 1 \end{array} \quad \begin{array}{cc|cc} 3 & 0 & 1 & -1 \\ -2 & -1 & 0 & 1 \end{array} \quad \begin{array}{cc|cc} 1 & 0 & 1/3 & -1/3 \\ 0 & 1 & 0 & 1 \end{array}$$

$$\begin{array}{cc|cc} 1 & 0 & 1/3 & -1/3 \\ 0 & 1 & -2/3 & 1/3 \end{array}$$

$$M^{-1}$$

$$A^{-1}:$$

$$\begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array} \quad \begin{array}{cc|cc} 1 & -1 & 1 & -1 \\ 2 & 3 & 0 & 1 \end{array} \quad \begin{array}{cc|cc} 1 & -1 & 1 & -1 \\ 0 & 5 & -2 & 3 \end{array} \quad \begin{array}{cc|cc} 1 & -1 & 1 & -1 \\ 0 & 1 & -2/5 & 3/5 \end{array} \quad \begin{array}{cc|cc} 1 & 0 & 3/5 & -2/5 \\ 0 & 1 & -2/5 & 3/5 \end{array}$$

$$D = M^{-1}A$$

$$\begin{bmatrix} 1/3 & -1/3 \\ -2/3 & -1/3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1/3 & -1/3 \\ -8/3 & -7/3 \end{bmatrix}}_D$$